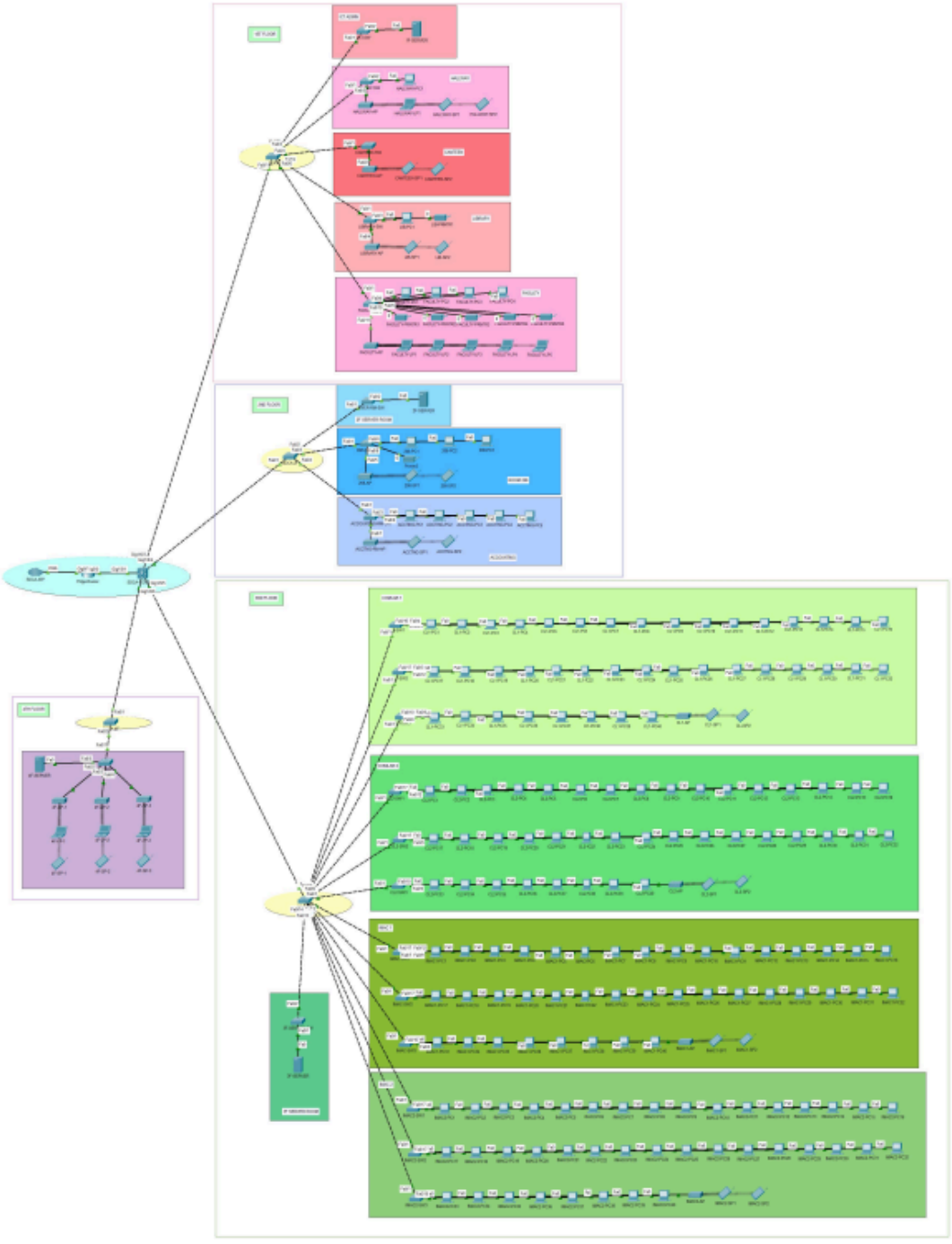
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NETWORKING MIDTERM

CAMPUS NETWORK DESIGN



The network design is a multi-floor campus network of SDCA that uses VLAN segmentation, centralized routing, DHCP, servers, and wireless connectivity to provide efficient communication. As shown in the topology, all floors are connected to a central device called the core switch (SDCA-CORE-SW), which acts as the main controller of the network. Each floor is assigned its own VLAN and subnet to organize traffic and improve performance. VLAN 10 is used for the first floor with the network 192.168.10.0/24, VLAN 20 is for the second floor with 192.168.20.0/24, VLAN 30 is for the third floor with 192.168.30.0/24, and VLAN 40 is for the fourth floor with 192.168.40.0/24. The core switch provides inter-VLAN routing, allowing devices from different floors to communicate with each other. It also acts as a DHCP server that automatically assigns IP addresses to devices using different pools per floor, where each pool includes a default gateway such as 192.168.10.1, 192.168.20.1, 192.168.30.1, and 192.168.40.1.

Each floor has access switches like hallway, ICT, faculty, and canteen switches that connect PCs, laptops, and access points. These switches assign devices to their respective VLANs and forward traffic to the core switch through trunk links. The network also includes wireless access points on each floor, with SSIDs such as SDCA-1F-AP, SDCA-2F-AP, SDCA-3F-AP, and SDCA-4F-AP1. Wireless devices like laptops and smartphones connect to these SSIDs and receive IP addresses from the DHCP server, allowing them to join the network. Each floor also has a server configured with a static IP address, such as 192.168.10.10, 192.168.20.10, 192.168.30.10, and 192.168.40.10, ensuring that services remain stable and always accessible. This setup follows a distributed server design, where each floor has its own server to reduce network load and improve performance.

The core switch is connected to an edge router using a small transit network with IP addresses 10.0.1.1 and 10.0.1.2. This connection is used only for routing traffic between the internal network and external networks. The edge router connects to the SDCA-ISP (cloud), which represents the school’s own ISP or backbone network. A default route is configured on the core switch to send unknown traffic to the router, and another default route is configured on the router to send traffic to the ISP. This allows devices inside the campus to communicate with external networks. Overall, the network works by organizing devices using VLANs, assigning IP addresses through DHCP, routing traffic using the core switch, and providing external connectivity through the edge router, creating a scalable, efficient, and realistic campus network system.